

Nebulous Drift: Project Overview & Scientific Report

1. Project Title:

Nebulous Drift - Astronomical Data Analysis & Celestial Observation

2. Team Members:

Nebulous Drift Group

3. Abstract / Project Summary:

Nebulous Drift is an amateur astronomy and data processing initiative focused on bridging observational astronomy with computational analysis. Our work involves analyzing astronomical imagery, processing multi-frame photography of celestial bodies (such as Saturn and deep-sky objects), and examining light intensity profiles using Python-based tools.

4. Tools & Technologies Used:

Programming Languages: Python (with AI assistance for code development and optimization)

Libraries & Frameworks: OpenCV, Astropy, Photutils, Matplotlib

Development Environment: Google Colab / Desktop Python

Observational Equipment: Reflector and refractor telescopes for target acquisition.

5. Methodology & Key Activities:

Image Stacking & Processing: Utilizing Python scripts developed with AI assistance to stack multi-frame captures of planets and stars, reducing atmospheric noise and enhancing surface details.

Light Curve Analysis: Generating and interpreting intensity profiles to study stellar or planetary data.

Citizen Science & Collaboration: Engaging in community space data projects and documenting celestial observations.

6. Future Goals:

Expanding our automated processing pipeline, publishing educational space content, and contributing further to amateur astronomical research